

AI Research Operating System

Product Architecture & Developer Brief

1. Overall Product Design Philosophy

The app must function as a **guided research workflow system**, not as a generic AI chat interface.

User journey must follow natural research progression:

1. Research Objective
2. Study Design
3. Sample Size
4. Proposal Development
5. Data Upload & Statistical Analysis
6. Results Compilation
7. Manuscript / Thesis Generation
8. Plagiarism & Compliance Check

The interface must be **menu-driven, structured, and modular**, not free-text dependent.

2. Core Modules (Phase 1 Launch)

Module 1: AI Research Assistant (Study Builder Engine)

Purpose: Guide the user from idea to structured methodology.

Functions:

- Accept research title or objective
- Suggest study design (RCT, cohort, cross-sectional, case-control, etc.)
- Define independent and dependent variables
- Generate inclusion/exclusion criteria
- Recommend sampling technique
- Suggest data collection structure

Output:

- Structured research blueprint

This is the brain of the system.

Module 2: Sample Size Calculator Engine

Purpose: Provide statistically correct sample size estimation.

Design Requirements:

- Menu-driven formula selection
- Based on study design
- Parametric and non-parametric study options

Core Formula Examples:

- Proportion studies
- Mean comparison
- Case-control
- Cohort
- Diagnostic studies
- Correlation studies

Output:

- Ideal calculated sample size
- Assumptions used (power, alpha, effect size)
- Editable confidence intervals

This module must be formula-driven, not AI-guess based.

Module 3: Statistical Analysis & Data Engine

Purpose: Convert uploaded Excel data into structured statistical outputs.

Workflow:

1. User uploads cleaned Excel dataset
2. User defines:
 - Independent variables
 - Dependent variables
3. System classifies data types:
 - Continuous
 - Categorical
 - Ordinal

Auto-Test Selection Logic:

- Normality check
- Parametric vs non-parametric selection
- Chi-square, t-test, ANOVA

- Mann-Whitney, Kruskal-Wallis
- Regression (linear, logistic)
- Correlation
- Diagnostic accuracy
- Basic survival analysis (future phase)

Advanced (Phase 2):

- Meta-analysis
- Linear discriminant analysis
- Forest plots
- ROC curves
- Box plots
- Kaplan-Meier

Important:

Currently menu-driven with statistician oversight.

App should automate test selection based on defined variables.

Output:

- Tables
- p-values
- Interpretation draft
- Graphs

Module 4: Proposal Generator (ICMR / Institutional Format)

Purpose: Structured proposal drafting.

Formats:

- ICMR
- Institutional Ethics Committee
- University PG thesis proposal

Sections:

- Background
- Aims & Objectives
- Methodology
- Sample size justification
- Statistical plan
- Data collection method

Output:

- Ready-to-submit proposal draft

Module 5: Thesis & Article Writing Engine

Purpose: Structured manuscript compilation.

Outputs:

- Thesis format (Chapter I–V)
- Journal-ready manuscript
- Vancouver referencing
- Abstract structured
- Table auto insertion

This must use structured data from previous modules, not free AI generation.

Module 6: Plagiarism & Compliance Engine

Purpose: Proactive integrity protection.

Features:

- Similarity scoring
- Highlight risk areas
- Suggest paraphrasing
- Citation cross-checking

Goal: Not reactive detection alone, but guided compliance.

3. User Workflow Order

Correct user navigation sequence:

- Step 1 → Enter Research Objective
- Step 2 → AI Study Builder
- Step 3 → Sample Size Calculation
- Step 4 → Proposal Generation
- Step 5 → Data Upload & Statistical Analysis
- Step 6 → Results Compilation
- Step 7 → Thesis / Article Writing
- Step 8 → Plagiarism & Compliance Review

The system must lock logical sequence unless user is advanced.

4. Application Design Structure

Design Type:

- Menu-driven architecture
- Modular dashboard
- Role-based access (Student / Researcher / Institution)
- Guided workflow with checkpoints
- Not chat-first UI

Interface Style:

- Structured panels
- Data tables
- Graph outputs
- Clear statistical menu selection

5. Services Required

Pre-Development Phase

- Product architecture blueprint
- UX workflow mapping
- Statistical formula validation by senior statistician
- Legal compliance framework
- IP registration planning

Development Phase

- Backend AI logic integration
- Statistical engine coding
- AWS cloud deployment
- Secure data encryption
- API integration for reference validation

Infrastructure

- AWS hosting
- Secure cloud storage
- Database scaling capability
- GDPR / Data compliance layer

Post-Launch Maintenance

- Bug fixes
- Statistical module updates

- Feature scaling (meta-analysis, forest plots)
- Security updates
- Server scaling

6. Total Modules at Launch

Phase 1: 6 Core Modules

Phase 2: Advanced Statistical & Meta-Analysis Layer

7. Positioning Summary

This is not:

- A writing AI
- A citation manager
- A statistics tool

This is a structured Research Operating System.

Design Approach

The application interface will follow a structured AI interaction model similar to leading conversational AI systems such as Claude.ai or ChatGPT, but adapted into a menu-driven, research-specific workflow.

Key Design Principles:

- Guided step-by-step navigation
- Structured input fields rather than open chat dependency
- Clear module progression
- Dashboard-based control panel
- Logical research sequence enforcement

The goal is not a free-text chatbot, but a structured research operating interface powered by AI reasoning.

Services Required

1. Pre-Development Phase

- Product architecture design and workflow mapping
- UX and UI blueprint development

- Statistical formula validation by certified statistician
- Legal and compliance framework review
- Data security planning
- IP registration and product documentation
- Cloud architecture planning

2. Infrastructure and AWS Requirements

- AWS cloud hosting environment
- Secure database management
- Scalable server architecture
- Encrypted data storage
- Role-based access control
- API integration for citation validation
- Backup and disaster recovery systems

AWS services likely required:

- EC2 (compute)
- RDS (database)
- S3 (storage)
- CloudWatch (monitoring)
- IAM (security control)

3. Post-Launch Maintenance

- Continuous bug fixes and system updates
- Statistical module upgrades
- Security patches and compliance updates
- Server scaling based on user growth
- Performance monitoring
- Feature expansion (advanced analysis, meta-analysis, graphs)
- Customer technical support

Maintenance must include periodic review by:

- AI engineer
- Backend developer
- Statistician

Summary

Total Launch Modules: 6 Core Modules

Design Style: Structured AI-powered workflow

Infrastructure: AWS-based secure cloud

Lifecycle: Pre-development → Build → Deploy → Maintain → Scale